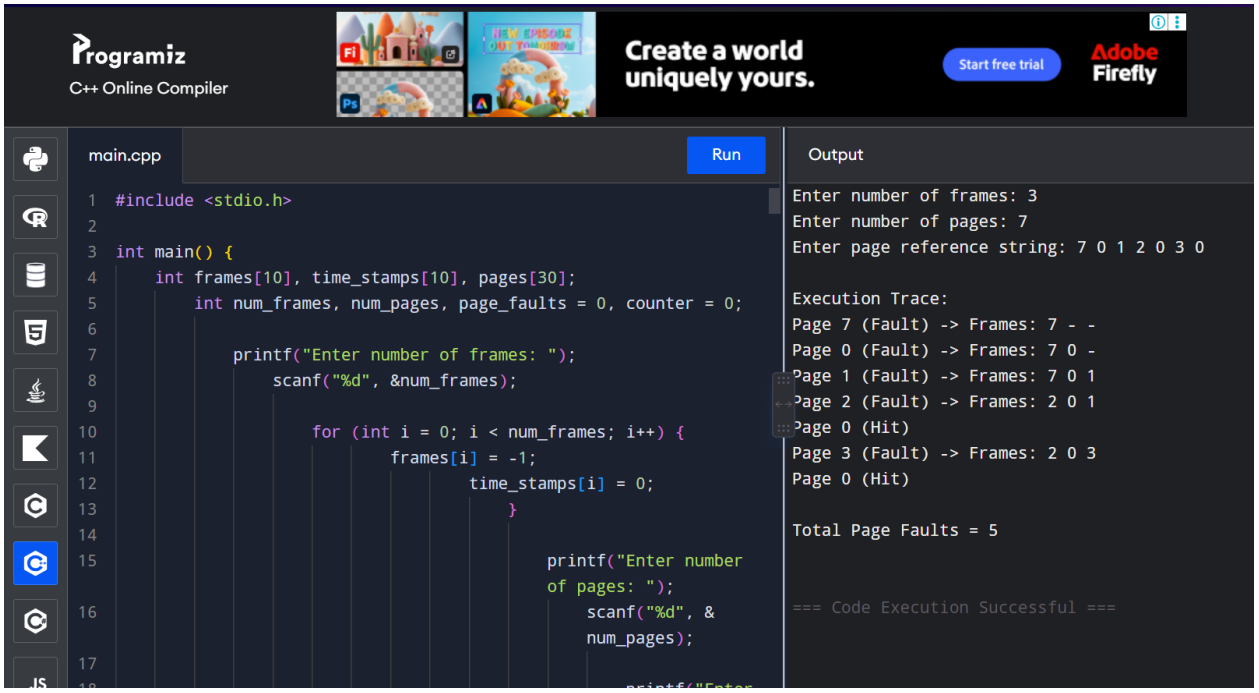
EXPERIMENT 26:

26. Construct a C program to simulate the Least Recently Used paging technique of memory management.

AIM:

To construct a C program to simulate the Least Recently Used (LRU) paging technique of memory management.



PSEUDO CODE :

BEGIN

PRINT "Enter number of frames:"

READ num_frames

DECLARE frames array of size num_frames

DECLARE time_stamps array of size num_frames

INITIALIZE frames with -1

INITIALIZE time_stamps with 0

```
PRINT "Enter number of pages:"
READ num_pages
DECLARE pages array of size num_pages

PRINT "Enter page reference string:"
FOR i FROM 0 TO num_pages - 1 DO
   READ pages[i]
END FOR

SET page_faults = 0
SET counter = 0

FOR i FROM 0 TO num_pages - 1 DO
   SET page = pages[i]
   SET page_found = 0
   INCREMENT counter

   // Search if page exists in frame
   FOR j FROM 0 TO num_frames - 1 DO
      IF frames[j] == page THEN
         SET page_found = 1
         SET time_stamps[j] = counter // Update last used timestamp
         BREAK
      END IF
   END FOR

   IF page_found == 0 THEN
      SET lru_index = 0
      SET empty_found = 0
```

```
        // Check for empty slot first
        FOR j FROM 0 TO num_frames - 1 DO
            IF frames[j] == -1 THEN
                SET lru_index = j
                SET empty_found = 1
                BREAK
            END IF
        END FOR

        // If no empty slot, find frame with minimum timestamp
        IF empty_found == 0 THEN
            SET min_time = time_stamps[0]
            SET lru_index = 0
            FOR j FROM 1 TO num_frames - 1 DO
                IF time_stamps[j] < min_time THEN
                    SET min_time = time_stamps[j]
                    SET lru_index = j
                END IF
            END FOR
        END IF

        SET frames[lru_index] = page
        SET time_stamps[lru_index] = counter
        INCREMENT page_faults

        PRINT "Page " + page + " (Fault) -> Frames: " + CURRENT_FRAMES
    ELSE
        PRINT "Page " + page + " (Hit)"
    END IF
END FOR
```

```
    PRINT "Total Page Faults: " + page_faults
END
```

CODE:

```
#include <stdio.h>



int main() {

    int frames[10],
time_stamps[10],
pages[30];

    int num_frames,
num_pages, page_faults
= 0, counter = 0;



    printf("Enter number
of frames: ");

    scanf("%d",
&num_frames);



    for (int i = 0; i <
num_frames; i++) {
```

```
        frames[i] = -1;
        time_stamps[i] = 0;
    }

    printf("Enter number
of pages: ");
    scanf("%d",
&num_pages);

    printf("Enter page
reference string: ");
    for (int i = 0; i <
num_pages; i++) {
        scanf("%d",
&pages[i]);
    }

    printf("\nExecution
Trace:\n");
```

```
    for (int i = 0; i <
num_pages; i++) {
        int page_found = 0;
        counter++;

        for (int j = 0; j <
num_frames; j++) {
            if (frames[j] ==
pages[i]) {
                page_found =
1;

time_stamps[j] =
counter;
                break;
            }
        }

        if (page_found ==
```

```
0) {
            int lru_index =
0;
            int empty_found
= 0;


            for (int j = 0; j <
num_frames; j++) {
                if (frames[j]
== -1) {
                    lru_index =
j;

empty_found = 1;
                    break;
                }
            }

            if
```

```
(!empty_found) {

            int min_time
= time_stamps[0];

            lru_index = 0;

            for (int j = 1; j
< num_frames; j++) {

                if
(time_stamps[j] <
min_time) {

                    min_time
= time_stamps[j];

                    lru_index
= j;

                }

            }

        }




frames[lru_index] =
pages[i];
```

```
time_stamps[lru_index]
= counter;
            page_faults++;


            printf("Page %d
(Fault) -> Frames: ",
pages[i]);
            for (int k = 0; k
< num_frames; k++) {
                if (frames[k]
!= -1)
                    printf("%d
", frames[k]);
                else
                    printf("- ");
            }
            printf("\n");
        } else {
```

```
            printf("Page %d
(Hit)\n", pages[i]);
        }
    }

    printf("\nTotal Page
Faults = %d\n",
page_faults);
    return 0;
}
```